



Published in final edited form as:

J Physiol. 2024 July ; 602(14): 3505–3518. doi:10.1113/JP286538.

Role of Na_v1.7 in Postganglionic Sympathetic Nerve Function in Human and Guinea Pig Arteries

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Abstract

Na_v1.7 plays a crucial role in inducing and conducting action potentials in pain-transducing sensory nociceptor fibres, suggesting that Na_v1.7 blockers could be effective non-opioid analgesics. While SCN9A is expressed in both sensory and autonomic neurons, its functional role in the autonomic system remains less established. Our single neuron rt-PCR analysis revealed that 82% of sympathetic neurons isolated from guinea pig stellate ganglia expressed Na_v1.7 mRNA, with Na_v1.3 being the only other tetrodotoxin-sensitive channel expressed in approximately 50% of neurons. We investigated the role of Na_v1.7 in conducting action potentials in postganglionic sympathetic nerves and in the sympathetic adrenergic contractions of blood vessels using selective Na_v1.7 inhibitors. Two highly selective Na_v1.7 blockers, GNE8493 and PF05089771, significantly inhibited postganglionic compound action potentials by approximately 70% (P<0.01), with residual activity being blocked by the Na_v1.3 inhibitor, ICA121431. Electrical field stimulation (EFS) induced rapid contractions in guinea pig isolated aorta, pulmonary arteries, and human isolated pulmonary arteries via stimulation of intrinsic nerves, which were inhibited by prazosin or the Na_v1 blocker tetrodotoxin. Our results demonstrated that blocking Na_v1.7 with GNE8493, PF05089771, or ST2262 abolished or strongly inhibited sympathetic adrenergic responses in guinea pigs and human vascular smooth muscle. These findings support the hypothesis that pharmacologically inhibiting Na_v1.7 could potentially reduce sympathetic and parasympathetic function in specific vascular beds and airways.

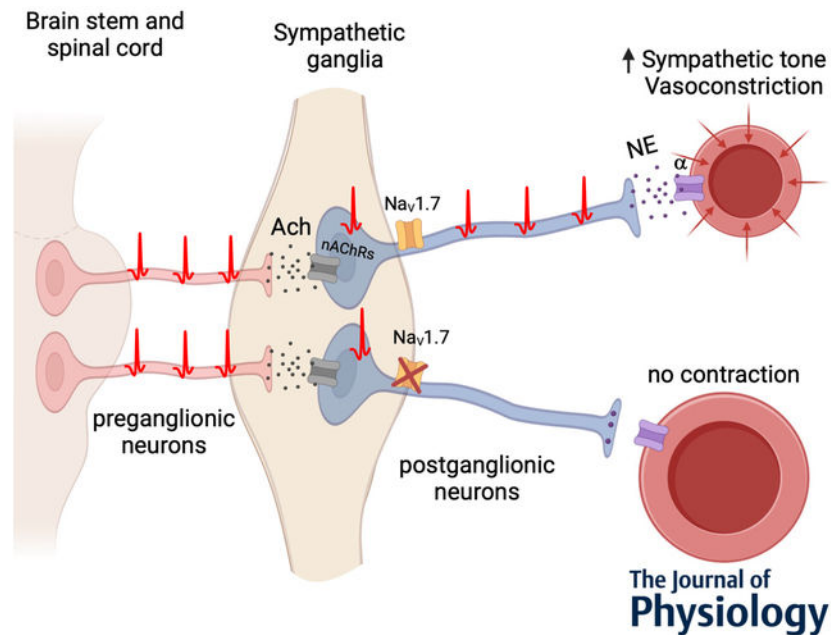
Graphical Abstract

Author Contributions

JSK, TSZ, DH, and BJU conceived and conducted the project and wrote the manuscript. JSK, SM, FR, and BJU contributed to the acquisition, analysis or interpretation of data. JSK performed single neuron cell picking and single neuron RT-PCR. SM performed tissue bath experiments. FR and MT performed post ganglionic nerve recording experiments. All authors contributed to and approved the final version of the manuscript. All person designated as authors qualify for authorship, and all those who qualify for authorship are listed.

Competing interest

The authors declare no conflict of interest. Authors from Genentech provided compounds and determined Na_v1 selectivity as well as being involved in the ideas underlying the study. They were not involved in final data analyses or production of figures.



Blocking Nav1.7 pharmacologically demonstrated a profound impact on sympathetic adrenergic responses in both guinea pigs and human vascular smooth muscle, leading to either complete abolition or strong inhibition. These findings support the hypothesis that pharmacologically inhibiting Nav1.7 could potentially reduce sympathetic functions in specific vascular beds. Ach: acetylcholine, nAChRs: nicotinic acetylcholine receptor, NE: norepinephrine, α : alpha-1 adrenoreceptor

Keywords

Voltage-gated sodium channels; nociceptors; autonomic nerves; pain

Introduction

There are nine voltage-gated sodium channel (VGSC) subtypes referred to as Nav1.1 – Nav1.9. With the exception of Nav1.9, these channels are responsible for the sodium current involved in the generation and conduction of action potentials (Catterall, 2012). Apart from Nav1.5 (mainly expressed in cardiac muscle) and Nav1.4 (in striated muscle), these channels are expressed in various nerve subtypes. The significance of Nav1 channels in regulating action potential generation and conduction in a nerve depends on the nerve subtype. This establishes the possibility of developing selective VGSC inhibitors to block the function of specific nerve subtypes.

The VGSC mRNA dominantly expressed in most small-diameter sensory C-fibre nociceptors is SCN9A, which translates to the α -subunit of Nav1.7 (Fukuoka et al., 2008; Dib-Hajj et al., 2010). These neurons also expressing the tetrodotoxin (TTX)-insensitive channels, Nav1.8 and 1.9. In contrast to nociceptors, Nav1.7 is rarely expressed in the central nervous system or large-diameter neurons within the somatosensory system. Based

on this, and motivated by the discovery that a loss-of-function mutation in $\text{Na}_V1.7$ leads to congenital insensitivity to pain (Cox et al., 2006; Dib-Hajj & Waxman, 2019), highly selective $\text{Na}_V1.7$ blockers have been discovered with the goal of developing peripheral-acting non-opioid analgesics (Lai et al., 2004; Payandeh & Hackos, 2018). Irrespective of therapeutic potential, selective VGSC inhibitors are well-suited to evaluate the role of Na_V1 subtypes in regulating the function of various nerve phenotypes.

Compared to the sensory nervous system, there have been relatively few studies on the VGSC subtypes most important in regulating the function of autonomic nerves. Postganglionic parasympathetic nerves regulating human and guinea pig (but not mice) airway smooth muscle express nearly exclusively $\text{Na}_V1.7$ mRNA (Kollarik *et al.*, 2022). Accordingly, selective $\text{Na}_V1.7$ inhibitors can abolish parasympathetic nerve-mediated cholinergic contractions of airway smooth muscle (Kollarik *et al.*, 2022). Studies have long noted that sympathetic neurons express $\text{Na}_V1.7$ (Toledo-Aral et al., 1997; Yang et al., 2004). $\text{Na}_V1.7$ in sympathetic and sensory nerves has been implicated in primary erythromelalgia (Cummins et al., 2004; Waxman & Dib-Hajj, 2005), but little remains known about the functional role of $\text{Na}_V1.7$ in sympathetic nerves.

In this study we evaluated the role of $\text{Na}_V1.7$ in postganglionic sympathetic nerve function. Selective $\text{Na}_V1.7$ blockers substantially inhibited sympathetic nerve-mediated adrenergic contraction of blood vessels isolated from human and guinea pig tissue. Moreover, these blockers inhibited action potential conduction in guinea pig sympathetic postganglionic axons.

Methods

Ethical approval

All animal experiments were approved by the Johns Hopkins School of Medicine Animal Care and Use Committee (IACUC; approval reference MO22M69). We adhered to all guidelines for ethical protocols and the care of experimental animals established by the National Institutes of Health (NIH). Human lungs were obtained from anonymous human organ donors through the International Institute for the Advancement of Medicine program (IIAM; Exton, PA). IIAM secures the appropriate authorization/consent from donors of non-transplantable organs and tissues for use in medical discovery and education by researchers. Given the complete anonymity of the donors, the Johns Hopkins Institutional Review Board (IRB) declared that studies on deidentified human lung tissue do not qualify as human subject research and granted us exemption from the requirement to obtain consent. Additionally, we confirm our understanding of the ethical principles under which the journal operates and assert that our work complies with the animal ethics checklist as outlined in the instructions to authors.

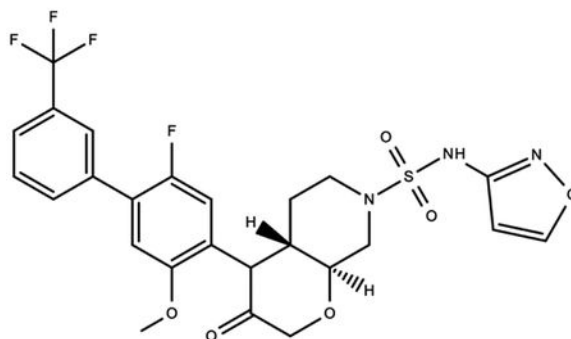
Animals

The animals were housed in an approved animal facility under a 12:12 h light/dark cycle with controlled temperature and humidity, in groups in cages providing unrestricted access to food and water (*ad libitum*). Male Hartley guinea pigs (100–200g) were sourced from

Hilltop Laboratory Animals (Scottsdale, PA). Euthanasia was carried out using 100% CO₂ (at a fill rate of 50% chamber vol/min) asphyxiation. We observed each guinea pig for lack of respiration and withdrawal reflex prior to tissue harvest.

Chemicals and reagents

All compounds and drugs used were freshly prepared from stock solutions stored at -20°C. Genentech Pharmaceuticals provided GNE8493, and SiteOne Pharmaceuticals provided ST2262. Tetrodotoxin was purchased from Alomone Labs, while ICA121431 and PF05089771 were purchased from Tocris. Atropine, barium chloride (BaCl₂), hexamethonium, indomethacin, N_ω-Nitro-L-arginine (LNNLA), phenylephrine, prazosin, and propranolol were purchased from Sigma-Aldrich. Chemicals were dissolved and prepared according to the manufacture's recommendations. On the day of the experiment, Kreb's bicarbonate buffer composed of (in mM): 118 sodium chloride, 5.4 KCl, 1.0 NaH₂PO₄, 1.2 MgSO₄, 1.9 CaCl₂, 25 NaHCO₃, and 11.1 glucose, was freshly prepared. The structure of ST2262 has previously been published (Pajouhesh *et al.*, 2020). The structure of GNE8493 is:



The Na_v1 subtype selectivity of GNE8493 was determined using methods described in (Safina *et al.*, 2021).

Dissociation of guinea pig stellate ganglia neurons

Right and left stellate ganglia were dissected and gently desheated. Multiple parallel slits (~1mm apart), perpendicular to the long axis, were made to isolated ganglia prior to incubation with enzyme buffer (2 mg collagenase and 2 mg dispase II in 1mL Ca²⁺ and Mg²⁺ free Hanks' balanced salt solution) for 60 min at 37 °C. Neurons were dissociated by trituration using three fire-polished glass Pasteur pipettes with decreasing tip pore size at the end of 30, 45, and 60min. The enzymatic solution containing the dissociated neurons (1 mL) was suspended in pre-warmed 10 mL Leibovitz's L-15 medium with 10% fetal bovine serum (FBS) and then centrifuged at 600g for 2 min. After an additional wash with L-15 medium (10% FBS), the cell suspension was transferred onto poly-D-lysine/laminin-coated coverslips and incubated at 37 °C for 2 h. Once the suspended neurons adhered to the coverslips for 2 h, the neuron-attached coverslips were flooded with the L-15 medium (10% FBS) and used within 8 h for cell picking.

Cell picking and single cell RT-PCR

Coverslips of dissociated neurons were constantly perfused with 1xPBS solution. A glass pipette (tip diameter 50–150µm) was pulled with a micropipette puller Model P-87, Sutter Instruments Co., Novato, CA, USA). A single cell was sucked into a glass pipette, and the pipette tip was subsequently broken in a PCR tube containing 1µl of RNase Inhibitor (RNaseOUT, Invitrogen, 2 U µl⁻¹). The cell was immediately snap frozen and stored in a -80°C freezer. One to four cells were collected from each coverslip. Additionally, a sample of the bath solution from the vicinity of a labeled neuron was collected from each coverslip for no-template experiments (bath control).

The tubes containing a single neuron, suspended in RNase OUT, were used to synthesize first strand cDNA using the SuperScript III First-Strand Synthesis System for RT-PCR (Cat#18 080 044, Invitrogen, Carlsbad, CA, USA) following the manufacturer's recommendations. In short, cell samples were defrosted, and then oligo(dT) and random hexamer primers (Roche Applied Bioscience) were added, and tubes incubated for 10 mins at 75°C. 22µl of the volume was reverse transcribed by adding Superscript III RT for cDNA synthesis, whereas water was added to the remaining sample, which was used as -RT control. 1.5 µl (cDNA, RNA control, or bath control) was used for PCR amplification of β -actin, Na ν 1.1, Na ν 1.2, Na ν 1.3, Na ν 1.6, and Na ν 1.9 by the HotStar Taq Polymerase Kit (Qiagen) according to the manufacturer's recommendations in a final volume of 20 µl. Primers targeting all exons specific for each guinea pig β -actin and Na $\nu\alpha$ subunits were designed using UCSC (University of California, Santa Cruz) Genome Browser and based on previously validated sequences on PubMed (Kocmalova et al., 2017; Ru et al., 2020). After an initial activation step at 95°C for 15 min, cDNAs were amplified with custom-synthesized primers (Sigma-Aldrich) (Table 1) by 50 cycles of amplification. The PCR program used consists of denaturation at 94°C for 30 s, annealing at 60°C for 30 s, and extension at 72°C for 1 min, followed by a final extension at 72°C for 10 min. PCR products were then visualized in ethidium bromide-stained 1.5% agarose gels. For full details on the single cell RT-PCR, please see our previous study (Kim *et al.*, 2023).

Selectivity of Na ν 1.7 blockers used in the functional studies below.

Conclusions regarding the role of Na ν 1.7 in functional studies on sympathetic nerves are based on the effect of selective Na ν 1.7 blockers. The data presented in Table 2 originate from patch-clamp studies and summarizes the relative selectivity of the VGSC blockers used in the functional studies.

Superior cervical ganglion (SCG) and post ganglionic nerve preparation

The sympathetic nerve with superior cervical ganglion (SCG) was dissected from a guinea pig. For nerve recording, only a very short segment (4mm~6mm) of the targeted postganglionic nerve proved suitable. To ensure complete drug delivery into the nerve, the associated sheath of the postganglionic nerve was carefully removed. Subsequently, the nerve with SCG was transferred to a custom-built water-jacketed tissue chamber lined with Sylgard and filled with 37°C oxygenated Krebs's solution (composition in mM: 118 NaCl, 5.4 KCl, 1 NaH₂PO₄, 1.2 MgSO₄, 1.9 CaCl₂, 25 NaHCO₃, and 11 dextrose, gassed with 95% O₂–5% CO₂).

Once set up in the tissue chamber, the postganglionic nerve and SCG ganglion were pinned in place to stabilize the tissue. The stimulating electrode tip was positioned abutting the SCG. The stimulating electrode was connected to an A-M Systems isolated pulse stimulator, model 2100. The nerve underwent stimulation with a square pulse (20 V, 0.1-ms pulse duration) every 10 seconds. These parameters were found to produce the largest reproducible CAPs in preliminary studies. The end of the postganglionic nerve was securely suctioned into a Krebs solution containing suction electrode (borosilicate glass capillary 1.0/0.75 OD/ID mm), connected to the head stage and AC amplifier (A-M Systems, Sequim, WA). A return silver–silver chloride wire electrode and earthed silver–silver chloride pellet were placed in the fluid of the tissue chamber. The recorded signal was then amplified (M1800; A-M Systems) and filtered (low cut-off: 0.1 Hz; high cut-off: 500 Hz). The compound action potential signal was displayed on an oscilloscope (Tektronix, TDS 2001C), and the amplitude (distance from peak to trough) of the compound action potential was quantified.

We added 300 μ M hexamethonium to the buffer solution to block any synaptic neurotransmission. In preliminary experiments we found this concentration abolished all compound action potentials (CAPs) evoked by specifically stimulating the preganglionic nerve. For the control response, the nerve was allowed to equilibrate for at least 30 minutes, during which time the compound action potential reached a steady state. If the compound action potential fluctuated during the equilibration period, it was deemed unsuitable for pharmacological analysis.

Different concentrations or combinations of Na v 1 blockers were tested. Each concentration was administered for 15 min, allowing the inhibitor to reach a quasi-steady state. The amplitude of the compound action potential at that 15-minute time point was recorded. Tetrodotoxin (1 μ M) was added to tissue at the end of the experiment as a positive control to confirm the penetration of the drugs into the nerve.

The percentage decrease in the amplitude of the compound action potential served as a rough approximation of the percentage of C-fibres silenced by the drug treatment. The inhibitors gain access to the de-sheathed section of nerve, reducing the compound action potential by blocking the conduction of the action potentials on its way to the recording electrode.

Tissue bath studies

Guinea pig abdominal aorta and pulmonary artery.—Abdominal aorta and pulmonary artery were removed and placed in Kreb's buffer equilibrated with 95% O $_2$ –5% CO $_2$ (pH 7.4), as described above. After removing extraneous tissue from the abdominal aorta and pulmonary artery, they were sectioned into 2–4 ring segments that were approximately 3–5mm in length. The preparations were connected to force-displacement transducers for measurements of circumferential isometric tension responses. They were equilibrated under 500 mg (abdominal aorta) or 1g (main pulmonary artery) resting tension for 60 minutes in water-jacketed tissue baths containing Kreb's buffer maintained at 37°C, with the buffer replaced at 15-minute intervals. The resting tensions were those, based on preliminary studies, that produced the maximal isometric contractions. Platinum

wires, connected to a Grass S48 stimulator, were positioned on either side of the tissue segments. Propranolol (1 μ M), indomethacin (3 μ M), atropine (1 μ M), LNNLA (100 μ M) were added to the buffer at the beginning of the experiment. This was done to isolate the alpha-adrenergic contractions. LNNLA was added to reduce any functional antagonism (relaxation) that may be caused by the release of endothelial nitric oxide, atropine was added to reduce any cholinergic responses that may lead to relaxations that are not inhibited by LNNLA, propranolol was added to reduced any beta-adrenoceptor mediated relaxation, and indomethacin was added to reduce the production of prostanoids.

The intrinsic nerves in the tissue were stimulated with electrical field stimulation (EFS) for 10 sec. The pulses were delivered to the tissue via the platinum wire electrodes from the Grass S48 stimulator after first being processed by a constant current Med-Laboratory Stimu-Splitter (Med-Laboratory Instruments, Fort Collins, CO). The square pulses were of 0.5msec duration with an amplitude of 12 V. The impulses were delivered at 32 Hz for 10 seconds trains on 100sec intervals. The electrical parameters were based on preliminary studies that determined the optimal setting to produce a maximal reproducible response that was abolished by 1 μ M tetrodotoxin. This resulted in rapid contraction in all tissues, which was entirely prevented with alpha adrenoceptor antagonists (1 μ M prazosin). After stable responses were obtained ~10–15 minutes, the Na ν 1.7 blocker (GNE8483 or PF05089771) was added until the reduction in response amplitude reached a steady state, at which time a 10-fold larger dose concentration was added. From data obtained, a cumulative concentration response curve was obtained. The contractions were quantified as the % of a maximal obtainable adrenergic contraction evoked by phenylephrine (10 $^{-5}$ M) followed by BaCl $_2$ (3x 10 $^{-2}$ M) at the end of the experiment. BaCl $_2$ was added to evoke a maximal obtainable contraction in smooth muscle.

Guinea pig trachea and human bronchi.—EFS-induced parasympathetic cholinergic contractions of guinea pig tracheal smooth muscle and human bronchial smooth muscle were evaluated as we described in our previous publication (Kocmalova et al., 2017).

Human pulmonary artery.—The tissue was immediately place in chilled Lactated Ringer solution and shipped overnight, arriving in our laboratory 12–24 hours later. Once received, the lungs were placed in 4 L chilled oxygenated Kreb's buffer (see above). Pulmonary artery (4–5 mm outer diameter) was dissected free, trimmed of extraneous tissue, and prepared for EFS-induced isometric study, as described for the guineapig abdominal aorta and pulmonary artery. For details on the tissue bath studies, see our previous study (Ellis & Udem, 1992).

Analysis

All statistical analyses were performed suing GraphPad Prism software (version 10.2.2). Data are expressed as mean $\pm$ S.D and $p < 0.05$ was considered statistically significant.

A nonlinear regression analysis of the concentration-response curves. From the data we derived the $-\log$ (M) EC $_{50} \pm$ SD. Paired Student's t-test was used to evaluate the statistical significance of the difference before and after drug treatment in human pulmonary artery data. Additionally, for multiple comparison we used the Kruskal-Wallis and Dunn's non-

parametric tests to assess the statistical significance between the delta of Control 1 vs. Control 2 values in the time controls compared to the delta between Control and drug-treated values.

Results

Single-neuron rt-PCR in Sympathetic Tissue

TTX effectively blocks all Na_V1 channels except $\text{Na}_V1.5$, 1.8 , and 1.9 . The conduction of action potentials in postganglionic sympathetic nerves as well as the sympathetic nerve mediated adrenergic contractions of guinea pig arteries are abolished by TTX (see results below). We therefore evaluated which of the TTX-sensitive VGSC α -subunits are most commonly expressed in sympathetic neurons isolated from the stellate ganglion. We found that 82% of the neurons (32/39) expressed $\text{Na}_V1.7$ mRNA. $\text{Na}_V1.1$ and 1.2 were not expressed in the neurons, whereas $\text{Na}_V1.3$ and $\text{Na}_V1.6$ were expressed in 54% (19/35) and 4% (1/28) of neurons, respectively; see figure 1.

Potency of VGSC blockers in an $\text{Na}_V1.7$ parasympathetic tissue bioassay

The potencies of VGSC blockers are based on IC_{50} values obtained from patch-clamp recordings, often conducted in transfected cells. These values can be strongly influenced by the concentration of extracellular sodium and, in the case of state-dependent blockers (e.g., arylsulfonamides, PF05089771, and GNE8493), by the holding resting membrane potential. We have previously published that $\text{Na}_V1.7$ is the only VGSC mRNA expressed in cholinergic parasympathetic neurons in human and guinea pig airway parasympathetic ganglia (Kocmalova et al., 2017). To evaluate the “functional potency” of a given $\text{Na}_V1.7$ blocker in a tissue assay, we carried out concentration-response curves in blocking parasympathetic nerve-mediate cholinergic contractions of human and guinea pig isolated airways.

The concentration-response curves for blocking guinea pig parasympathetic responses are illustrated in figure 2A and the responses in human isolated bronchi are shown in figure 2B. In our recently published study, we found the $-\log(M) \text{EC}_{50}$ for PF05089771 to be 5.6 ± 0.2 with the response nearly abolished at $10 \mu\text{M}$ of the compound (Kollarik *et al.*, 2022). The ST2262 compound was not used in guinea pig tissues as it has been reported to have relatively little affinity for rat $\text{Na}_V1.7$, and this was supported by our finding that the cholinergic responses in the guinea pig were only modestly inhibited at $10 \mu\text{M}$ (data not shown).

Action Potential Conduction in Sympathetic Tissue

We isolated the superior cervical ganglion along with a segment of the sympathetic trunk rostral to the ganglion to record the compound action potential (CAP) in the post-ganglionic neurons. Figure 3 shows a typical CAP in the postganglionic sympathetic nerve trunk. We found that, depending on the experiment the CAP may comprise 1, 2, or 3 peaks. The example in figure 3 shows 3 peaks; all peaks always reflected axons conducting action potentials at $<1 \text{ m/sec}$. The bar graphs show that the response of the CAP evoked by 1 stimulus/10 sec for 60 minutes is stable over time ($n=3$). Figure 4 shows that the CAP

is entirely blocked by TTX and strongly inhibited by the selective $\text{Na}_V1.7$ antagonists GNE8493 and PF055089771 ($n=6$, $n=5$, $n=5$, respectively). Neither of the selective $\text{Na}_V1.7$ blockers, at the maximum concentration evaluated abolished the response. Based on aforementioned gene expression results, we hypothesized that these residual responses may be secondary to $\text{Na}_V1.3$ activation. We noted that adding ICA-121341, a compound that effectively blocks $\text{Na}_V1.3$ (see table 2), to the response remaining in the presence of $10 \mu\text{M}$ of PF055089771 abolished the response (figure 4 C).

Guinea pig aorta

We next evaluated the influence of $\text{Na}_V1.7$ blockade on the ability of the sympathetic nerves to regulate vascular smooth muscle tone. Despite evidence suggesting the expression of $\text{Na}_V1.7$ in certain vascular myocytes, the ability of adrenoceptor activation to contract aortic smooth muscle was not influenced by TTX-sensitive VGSCs. We carried out cumulative concentration-response curves for phenylephrine-induced contractions of the arteries in the absence and presence of TTX ($1 \mu\text{M}$) and found the $-\log(\text{M})\text{EC}_{50\text{s}}$ to be 5.5 ± 0.1 and 5.6 ± 0.2 , respectively ($P>0.1$). The maximum adrenergic contraction relative to BaCl_2 was $24 \pm 4\%$ and $22 \pm 1\%$, respectively ($P>0.1$).

Using an electrical field stimulation assay, we noted that the guinea pig isolated aorta contracted to 60 sec of an 8Hz train of EFS with a rapid contraction of the vascular smooth muscle. These responses were abolished by the α_1 -adrenoceptor blocker prazosin (data not shown). These adrenergic contractions averaged about 15–30% of a maximal adrenergic contraction evoked by phenylephrine added at the end of the experiment.

TTX ($n=5$), GNE8493 ($n=7$), and PF055089771 ($n=5$) were each capable of blocking the EFS-induced contractions of the aorta with potencies similar to that seen in the $\text{Na}_V1.7$ airway parasympathetic nerve assay (see figure 5A).

Guinea pig pulmonary artery

We also evaluated the influence of Na_V1 blockers in the adrenergic, prazosin-sensitive, contractions of guinea pig isolated pulmonary arteries using the same experimental design. The EFS-evoked stronger contraction in the pulmonary arteries that averaged about 40% of the maximum adrenergic response evoked by phenylephrine. TTX and the two selective $\text{Na}_V1.7$ blockers effectively inhibited the sympathetic adrenergic responses with potencies comparable to those observed in both the guinea pig trachea (parasympathetic) and aorta (sympathetic) (figure 5). TTX ($n=9$) was capable of abolishing the responses in the pulmonary arteries, although a small nerve-mediated response remained in the presence of the maximum concentration tested for the two selective $\text{Na}_V1.7$ blockers (GNE 8493, $n=4$ and PF055089771, $n=12$). Consistent with aforementioned gene expression results, ICA-121341 blocked the remaining response in the presence of $10 \mu\text{M}$ PF055089771 (figure 5B)

Human Pulmonary Artery

Pulmonary arteries isolated from human lung donors were studied in the same fashion as guinea pig arteries. The pulmonary arteries responded to EFS with contractions

that averaged ~10%-20% of the phenylephrine maximum. The contraction to EFS were prevented by prazosin (1 μ M). The responses to EFS were relatively stable over time, but in some cases, responses faded over longer periods of time. We therefore limited our analysis to a control response vs. a response in the presence of a single maximally effective concentration of a selective Na_v1.7 blocker, based on the results with human bronchi (parasympathetic). We found that GNE8483 (n=6), ST2262 (n=5), and PF05089771 (n=4) each nearly abolished the sympathetic nerve-mediated adrenergic contraction of the human pulmonary arterial smooth muscle (figure 6). As with the guinea pig arteries, we found no evidence that blocking TTX-sensitive channels altered the response of the smooth muscle to phenylephrine. The response of the arteries to phenylephrine (10 μ M) averaged 48 $\pm$ 9% and 54 $\pm$ 9% of the BaCl₂ maximum contraction in the absence and presence of TTX, respectively (n = 4, paired T-test, P = 0.15).

Discussion

This study supports the hypothesis that blocking Na_v1.7 channels in postganglionic sympathetic nerves can inhibit impulse conduction, leading to a substantial interference with adrenergic regulation of vascular smooth muscle tone in both guinea pig and human blood vessels. Selective Na_v1.7 blockers inhibit these responses with potencies similar to those seen for the inhibition of Na_v1.7-dependent parasympathetic nerve-mediate cholinergic contractions of human and guinea pig isolated airways. This hypothesis finds support from single-cell rtPCR analysis showing that Na_v1.7 is the most prominent Na_v1 expressed by the vast majority of sympathetic neurons isolated from the stellate ganglia as well as prior published evidence in the parasympathetic nervous system (Kocmalova *et al.*, 2017). It is further reinforced by the inhibition of action potential conduction in guinea pig post-ganglionic nerves with selective Na_v1.7 blockers, and by the ability of selective Na_v1.7 blockers to strongly inhibit nerve-evoked adrenergic contractions of both guinea pig systemic and pulmonary arteries, as well as human arteries.

While there is some evidence suggesting that Na_v1.7 is expressed in vascular smooth muscle (Rice *et al.*, 2015), our findings reveal that blocking Na_v1.7 channels had no effect on phenylephrine-induced contractions of the isolated vessels. This supports the conclusion that the blockade of EFS-induced adrenergic contractions was due to inhibition of norepinephrine released from sympathetic nerves in the tissues, rather than a direct effect on smooth muscle. It is essential to note that our study focused only on the isolated aorta from the guinea pig and isolated pulmonary arteries from guinea pigs and human lung donors. Thus, it would be imprudent to extrapolate these findings to all vascular beds. Nevertheless, the single-neuron rt-PCR provides evidence that nearly all sympathetic neurons in the guinea pig stellate ganglia express Na_v1.7 mRNA, with relatively little expression of other TTX-sensitive Na_v1s. The expression of Na_v1.7 in sympathetic neurons has previously been noted by others (Toledo-Aral *et al.*, 1997; Yang *et al.*, 2004).

The potency of TTX and Na_v1.7 blockers in inhibiting action potential conduction in postganglionic nerves was similar to what we have recently published regarding action potential conduction in vagal C-fibres innervating human airways (Kollarik *et al.*, 2022). The amplitude of the CAP in the postganglionic sympathetic nerve is a function of the

number of axons conducting a stimulus-evoked action potential. The CAP we studied reflects action potentials being conducted at a relatively slow velocity (<1 m/s), consistent with postganglionic nerves. TTX's ability to essentially abolish these potentials in the sympathetic nerves indicates that blocking TTX-sensitive channels is sufficient to block action potential conduction in nearly all the axons in the nerve trunk. We found that despite administering the two selective Na_v1.7 blockers at concentration capable of essentially abolishing the sympathetic-nerve evoked contraction of the blood vessels and Na_v1.7-dependent parasympathetic contractions of airway smooth muscle, they inhibited the CAP by ~60–80%. This suggests that a subset of axons is capable of conducting action potentials even in the absence of functional Na_v1.7 channels. The mRNA expression data point towards Na_v1.3 as the additional TTX-sensitive channel. Consistent with this, ICA-121341, a compound capable of blocking Na_v1.3 (McCormack et al., 2013), inhibited the CAP observed after Na_v1.7 blockade.

With respect to the arteries, whereas the aortas failed to respond in the presence of the Na_v1.7 blockers, the response of both guinea pig and human pulmonary arteries, similar to the CAP study, was not abolished by the selective Na_v1.7 blockers. This implies that either all of the Na_v1.7 channels were not blocked by the largest concentration of the blockers tested, or another Na_v (e.g., Na_v1.3) may allow for a small sympathetic response. We favor the later hypothesis because the largest concentrations of the Na_v1.7 blockers were capable of abolishing responses in both the airways and guinea pig aorta. Moreover, in keeping with the CAP study, adding ICA-121341 on top of 10 μM PF05089771 nearly abolished the EFS-induced responses of guinea pulmonary arteries.

These findings collectively suggest that an effective Na_v1.7 blocker could potentially lead to hypotensive effects. This prediction has recently been validated in guinea pigs and rats, where a relatively selective Na_v1.7 blocker was associated with decreases in blood pressure (Yu *et al.*, 2021; Stone *et al.*, 2015). Moreover, larger doses of a selective Na_v1.7 blocker have been associated with hypotensive responses in a human safety and tolerability trial (Rothenberg et al., 2019).

Both gain-of-function and loss-of-function mutations have been associated with various pathologies including erythromelalgia and congenital insensitivity to pain (Cummins et al., 2004; Waxman & Dib-Hajj, 2005; Dib-Hajj et al., 2013). Individual with loss-of-function mutation in Na_v1.7 have a loss in pain sensation as well as defects in olfaction (Cox et al., 2006; Weiss et al., 2011; Dib-Hajj et al., 2013). The result here, along with studies examining the parasympathetic system (Kocmalova et al., 2017), which highlights the profound effect of pharmacologically blocking Na_v1.7 on at least some sympathetic and parasympathetic responses, may motivate additional studies into the role Na_v1.7 mutation in causing other, perhaps more subtle, physiological effects that are a consequence of changes in autonomic nerve activity.

Funding

This work as supported by NIH NHLBI R35HL155671, F32HL170490, and by Genentech Inc.

Biography



Joyce S. Kim is a postdoctoral fellow at Johns Hopkins School of Medicine in Dr. Brad Undem's lab. She earned her Ph.D. from Penn State College of Medicine under the supervision of Dr. Marc Kaufman, conducting research on the autonomic regulation of the exercise pressor reflex in skeletal muscles. Dr. Kim's current research focuses on investigating the pharmacology and mechanisms that drive vagal afferent excitability in both healthy conditions and in the context of airway inflammatory diseases.

Data availability statement

All data in this study are available upon reasonable request to the corresponding author.

References

- Alexander SPH, Mathie AA, Peters JA, Veale EL, Striessnig J, Kelly E, Armstrong JF, Faccenda E, Harding SD, Davies JA, Aldrich RW, Attali B, Baggetta AM, Becirovic E, Biel M, Bill RM, Caceres AI, Catterall WA, Conner AC, Davies P, De Clerq K, Delling M, Di Virgilio F, Falzoni S, Fenske S, Fortuny-Gomez A, Fountain S, George C, Goldstein SAN, Grimm C, Grissmer S, Ha K, Hammelmann V, Hanukoglu I, Hu M, Ijzerman AP, Jabba SV, Jarvis M, Jensen AA, Jordt SE, Kaczmarek LK, Kellenberger S, Kennedy C, King B, Kitchen P, Liu Q, Lynch JW, Meades J, Mehlfeld V, Nicke A, Offermanns S, Perez-Reyes E, Plant LD, Rash L, Ren D, Salman MM, Sieghart W, Sivilotti LG, Smart TG, Snutch TP, Tian J, Trimmer JS, Van den Eynde C, Vriens J, Wei AD, Winn BT, Wulff H, Xu H, Yang F, Fang W, Yue L, Zhang X & Zhu M. (2023). The Concise Guide to PHARMACOLOGY 2023/24: Ion channels. *Br J Pharmacol* 180 Suppl 2, S145–S222. [PubMed: 38123150]
- Alexandrou AJ, Brown AR, Chapman ML, Estacion M, Turner J, Mis MA, Wilbrey A, Payne EC, Gutteridge A, Cox PJ, Doyle R, Printzenhoff D, Lin Z, Marron BE, West C, Swain NA, Storer RI, Stuppel PA, Castle NA, Hounshell JA, Rivara M, Randall A, Dib-Hajj SD, Krafte D, Waxman SG, Patel MK, Butt RP & Stevens EB. (2016). Subtype-Selective Small Molecule Inhibitors Reveal a Fundamental Role for Nav1.7 in Nociceptor Electrogenesis, Axonal Conduction and Presynaptic Release. *PLoS One* 11, e0152405. [PubMed: 27050761]
- Catterall WA. (2012). Voltage-gated sodium channels at 60: structure, function and pathophysiology. *J Physiol* 590, 2577–2589. [PubMed: 22473783]
- Cox JJ, Reimann F, Nicholas AK, Thornton G, Roberts E, Springell K, Karbani G, Jafri H, Mannan J, Raashid Y, Al-Gazali L, Hamamy H, Valente EM, Gorman S, Williams R, McHale DP, Wood JN, Gribble FM & Woods CG. (2006). An SCN9A channelopathy causes congenital inability to experience pain. *Nature* 444, 894–898. [PubMed: 17167479]
- Cummins TR, Dib-Hajj SD & Waxman SG. (2004). Electrophysiological properties of mutant Nav1.7 sodium channels in a painful inherited neuropathy. *J Neurosci* 24, 8232–8236. [PubMed: 15385606]
- Dib-Hajj SD, Cummins TR, Black JA & Waxman SG. (2010). Sodium channels in normal and pathological pain. *Annu Rev Neurosci* 33, 325–347. [PubMed: 20367448]
- Dib-Hajj SD & Waxman SG. (2019). Sodium Channels in Human Pain Disorders: Genetics and Pharmacogenomics. *Annu Rev Neurosci* 42, 87–106. [PubMed: 30702961]
- Dib-Hajj SD, Yang Y, Black JA & Waxman SG. (2013). The Na(V)1.7 sodium channel: from molecule to man. *Nat Rev Neurosci* 14, 49–62. [PubMed: 23232607]

- Ellis JL & Undem BJ. (1992). Antigen-induced enhancement of noncholinergic contractile responses to vagus nerve and electrical field stimulation in guinea pig isolated trachea. *J Pharmacol Exp Ther* 262, 646–653. [PubMed: 1323657]
- Fukuoka T, Kobayashi K, Yamanaka H, Obata K, Dai Y & Noguchi K. (2008). Comparative study of the distribution of the alpha-subunits of voltage-gated sodium channels in normal and axotomized rat dorsal root ganglion neurons. *J Comp Neurol* 510, 188–206. [PubMed: 18615542]
- Kim JS, Sun H, Meeker S & Undem BJ. (2023). Role of Na V 1.9 in inflammatory mediator-induced activation of mouse airway vagal C-fibres. *J Physiol* 601, 1139–1150. [PubMed: 36750759]
- Kocmalova M, Kollarik M, Canning BJ, Ru F, Adam Herbstsomer R, Meeker S, Fonquerna S, Aparici M, Miralpeix M, Chi XX, Li B, Wilenkin B, McDermott J, Nisenbaum E, Krajewski JL & Undem BJ. (2017). Control of Neurotransmission by Nav1.7 in Human, Guinea Pig, and Mouse Airway Parasympathetic Nerves. *J Pharmacol Exp Ther* 361, 172–180. [PubMed: 28138042]
- Kollarik M, Ru F, Pavelkova N, Mulcahy J, Hunter J & Undem BJ. (2022). Role of Na V 1.7 in action potential conduction along human bronchial vagal afferent C-fibres. *Br J Pharmacol*. 2022 Jan;179(2):242–251. doi: 10.1111/bph.15686. Epub 2021 Nov 8. *Br J Pharmacol* **179**, 242–251. [PubMed: 34634134]
- Lai J, Porreca F, Hunter JC & Gold MS. (2004). Voltage-gated sodium channels and hyperalgesia. *Annu Rev Pharmacol Toxicol* 44, 371–397. [PubMed: 14744251]
- McCormack K, Santos S, Chapman ML, Krafte DS, Marron BE, West CW, Krambis MJ, Antonio BM, Zellmer SG, Printzenhoff D, Padilla KM, Lin Z, Wagoner PK, Swain NA, Stuppel PA, de Groot M, Butt RP & Castle NA. (2013). Voltage sensor interaction site for selective small molecule inhibitors of voltage-gated sodium channels. *Proc Natl Acad Sci U S A* 110, E2724–2732. [PubMed: 23818614]
- Pajouhesh H, Beckley JT, Delwig A, Hajare HS, Luu G, Monteleone D, Zhou X, Ligutti J, Amagasu S, Moyer BD, Yeomans DC, Du Bois J & Mulcahy JV. (2020). Discovery of a selective, state-independent inhibitor of Na. *Sci Rep* 10, 14791. [PubMed: 32908170]
- Payandeh J & Hackos DH. (2018). Selective Ligands and Drug Discovery Targeting the Voltage-Gated Sodium Channel Nav1.7. *Handb Exp Pharmacol* 246, 271–306. [PubMed: 29532179]
- Rice FL, Albrecht PJ, Wymer JP, Black JA, Merkies IS, Faber CG & Waxman SG. (2015). Sodium channel Nav1.7 in vascular myocytes, endothelium, and innervating axons in human skin. *Mol Pain* 11, 26. [PubMed: 25957174]
- Rothenberg ME, Tagen M, Chang JH, Boyce-Rustay J, Friesenhahn M, Hackos DH, Hains A, Sutherland D, Ward M & Cho W. (2019). Safety, Tolerability, and Pharmacokinetics of GDC-0276, a Novel Na. *Clin Drug Investig* 39, 873–887.
- Ru F, Pavelkova N, Krajewski JL, McDermott JS, Undem BJ & Kollarik M. (2020). Stimulus intensity-dependent recruitment of Na. *Am J Physiol Gastrointest Liver Physiol* 319, G443–G453. [PubMed: 32726130]
- Safina BS, McKerrall SJ, Sun S, Chen CA, Chowdhury S, Jia Q, Li J, Zenova AY, Andrez JC, Bankar G, Bergeron P, Chang JH, Chang E, Chen J, Dean R, Decker SM, DiPasquale A, Focken T, Hemeon I, Khakh K, Kim A, Kwan R, Lindgren A, Lin S, Maher J, Mezeyova J, Misner D, Nelkenbrecher K, Pang J, Reese R, Shields SD, Sojo L, Sheng T, Verschoof H, Waldbrook M, Wilson MS, Xie Z, Young C, Zabka TS, Hackos DH, Ortwine DF, White AD, Johnson JP, Robinette CL, Dehnhardt CM, Cohen CJ & Sutherland DP. (2021). Discovery of Acyl-sulfonamide Na. *J Med Chem* 64, 2953–2966. [PubMed: 33682420]
- Stone AJ, Copp SW & Kaufman MP. (2015). Role played by NaV 1.7 channels on thin-fiber muscle afferents in transmitting the exercise pressor reflex. *Am J Physiol Regul Integr Comp Physiol* 309, R1301–R1308. [PubMed: 26310938]
- Toledo-Aral JJ, Moss BL, He ZJ, Koszowski AG, Whisenand T, Levinson SR, Wolf JJ, Silos-Santiago I, Haleboua S & Mandel G. (1997). Identification of PN1, a predominant voltage-dependent sodium channel expressed principally in peripheral neurons. *Proc Natl Acad Sci U S A* 94, 1527–1532. [PubMed: 9037087]
- Waxman SG & Dib-Hajj SD. (2005). Erythromelalgia: a hereditary pain syndrome enters the molecular era. *Ann Neurol* 57, 785–788. [PubMed: 15929046]

- Weiss J, Pyrski M, Jacobi E, Bufe B, Willnecker V, Schick B, Zizzari P, Gossage SJ, Greer CA, Leinders-Zufall T, Woods CG, Wood JN & Zufall F. (2011). Loss-of-function mutations in sodium channel Nav1.7 cause anosmia. *Nature* 472, 186–190. [PubMed: 21441906]
- Yang Y, Wang Y, Li S, Xu Z, Li H, Ma L, Fan J, Bu D, Liu B, Fan Z, Wu G, Jin J, Ding B, Zhu X & Shen Y. (2004). Mutations in SCN9A, encoding a sodium channel alpha subunit, in patients with primary erythralgia. *J Med Genet* 41, 171–174. [PubMed: 14985375]
- Yu L, Tsuji K, Ujihara I, Liu Q, Pavelkova N, Tsujimura T, Inoue M, Meeker S, Nisenbaum E, McDermott JS, Krajewski J, Undem BJ, Kollarik M & Canning BJ. (2021). Antitussive effects of Na. *Eur J Pharmacol* 907, 174192. [PubMed: 34010618]

Key points

- 82% of sympathetic neurons isolated from the stellate ganglion predominantly express Na_v1.7 mRNA.
- Na_v1.7 blockers inhibit action potential conduction in postganglionic sympathetic nerves
- Na_v1.7 blockade substantially inhibits sympathetic nerve-mediated adrenergic contractions in human and guinea pig blood vessels.
- Pharmacologically blocking Na_v1.7 profoundly affects sympathetic and parasympathetic responses in addition to sensory fibres, prompting exploration into the broader physiological consequences of Na_v1.7 mutations on autonomic nerve activity.

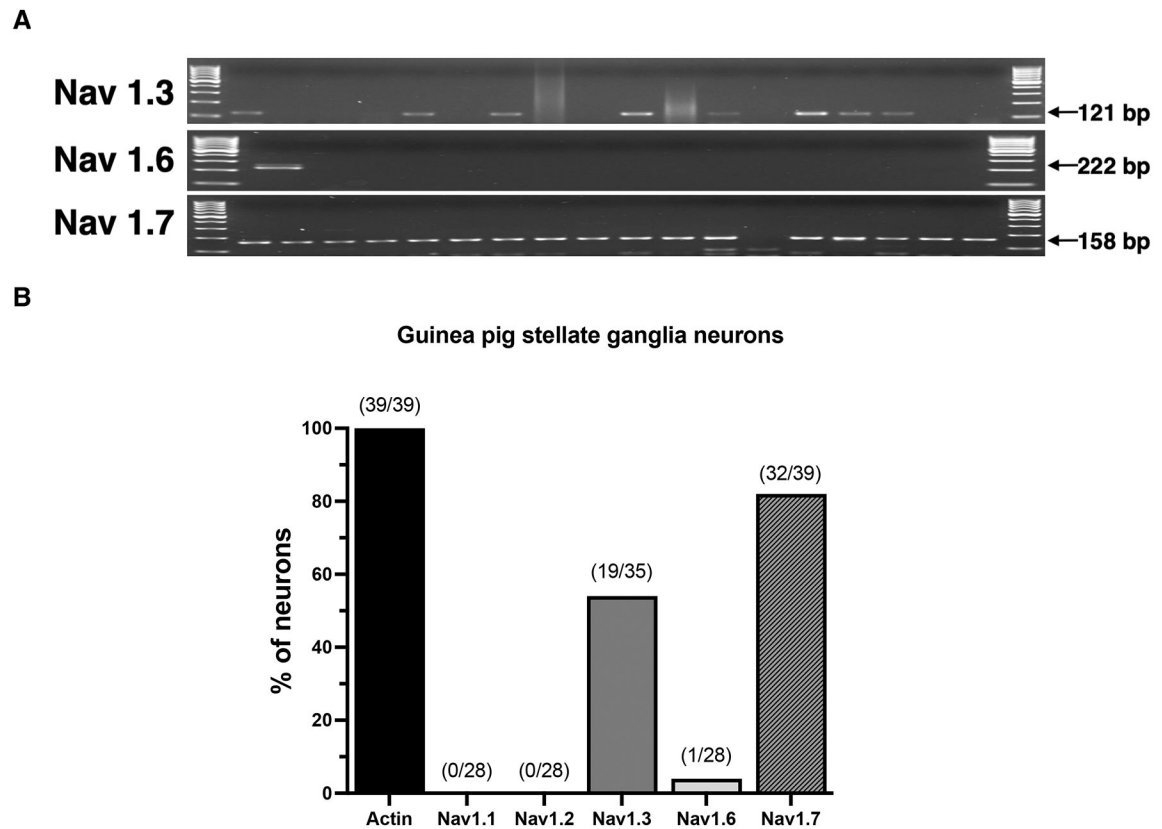
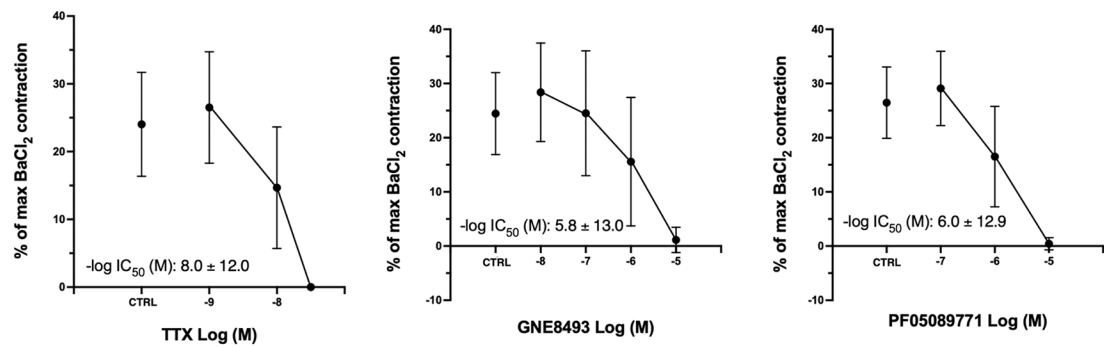


Figure 1.

Single neurons rt-PCR for Na_V1 subtypes obtained from sympathetic neurons dissociated from guinea pig stellate ganglia. A total of 39 actin-positive neurons (positive-control) were evaluated, obtained from 5 ganglia. ((A) representative results revealing $\text{Na}_V1.7$ (n=18 cells), $\text{Na}_V1.3$ (n=18 cell), and $\text{Na}_V1.6$ (n=10 cells) expression. (B) a summary of results from all neurons studied.

A (Guinea pig trachea)



B (Human bronchi)

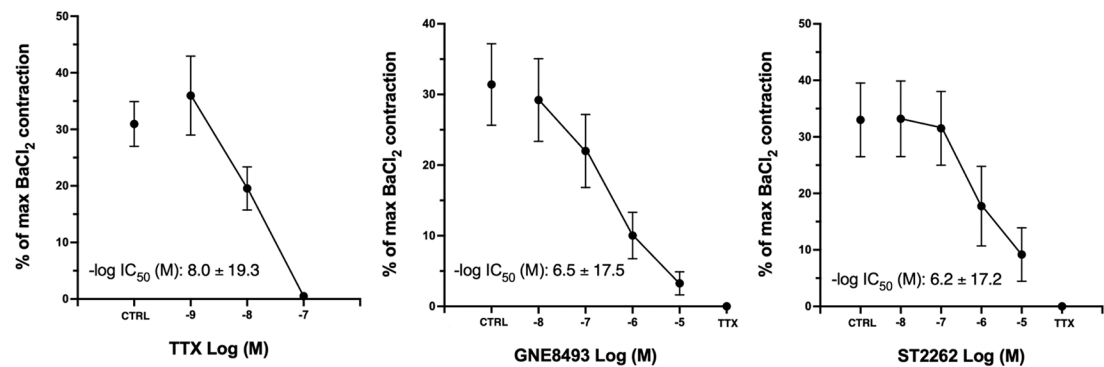


Figure 2.

Concentration-inhibition curves for Na_v1 blockers inhibiting EFS-induced cholinergic contractions of (A) guinea pig isolated trachea (TTX, n=16; GNE8493, n=11; and PF05089771, n=9) and (B) human isolated bronchi (TTX, n=21; GNE8493, n=10; and ST2262, n=6). The contractions are expressed as a % of the maximal obtainable contraction evoked by barium chloride (BaCl₂) added at the end of the experiment. The data present the mean $\pm$ SD of 6–8 experiments. We have previously noted that the contractions are dependent on Na_v1.7 channel activation (Kocmalova et al., 2017).

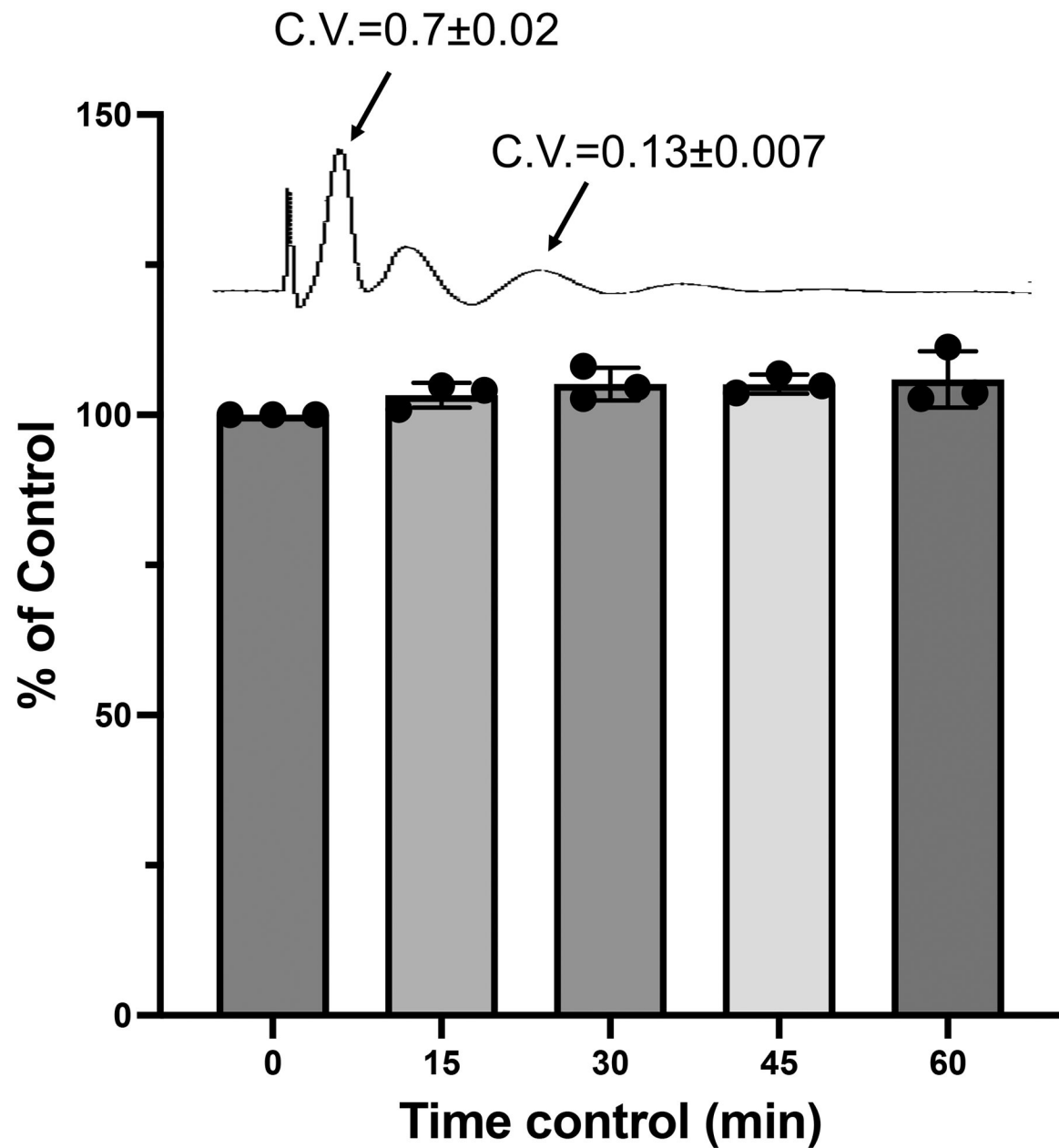


Figure 3.

Representative CAP is the postganglionic sympathetic nerve trunk caused by 20V 0.1 msec square pulse stimulation. We found the CAP may comprise 1, 2, or 3 peaks depending on the experiment. The exemplary trace shows 3 peaks, but all peaks always reflected axons conducting action potentials at <1 m/sec. The peak-trough amplitude was stable over the course of the experiments (data reflect the peak response as a % of that response taken at time 0, an $n=3$ separate ganglion experiments). The bar graphs show the average response of the CAP evoked by 1 stimulus/10 sec for 60 minutes. The data represent the mean $\pm$ SD.

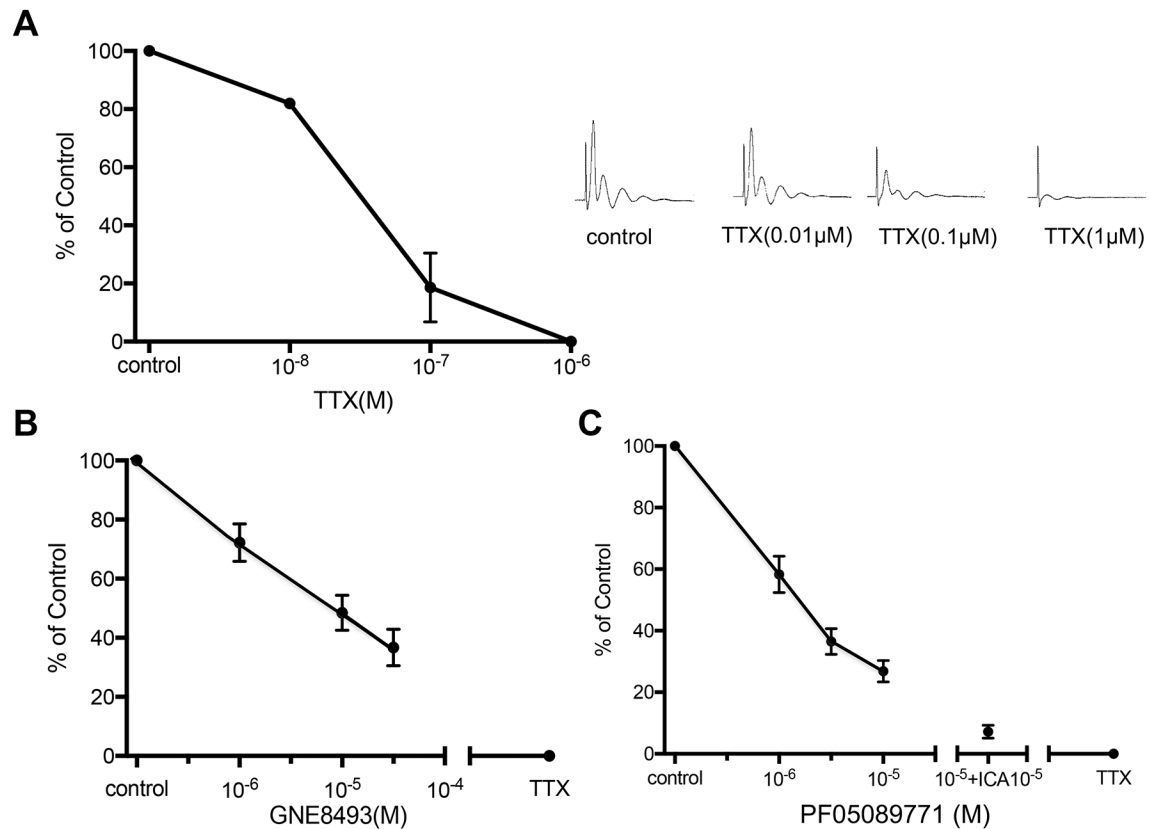


Figure 4.

Concentration-inhibition curves for (A) TTX-, (B) GNE8483-, and (C) PF05089771-induced inhibition of the CAP in postganglionic sympathetic nerves ($n=6$, $n=5$, $n=5$, respectively). Note that in C. the response left over in the presence of 10 μ M PF05089771 was essentially blocked the addition of ICA-121341. Top right shows a representative tracing of CAP presence of TTX. The data represent the mean $\pm$ SD of $n = 5$ or 6 different guinea pig experiments. The exemplary trace of a representative experiment illustrates the concentration-dependent inhibition of CAP with TTX.

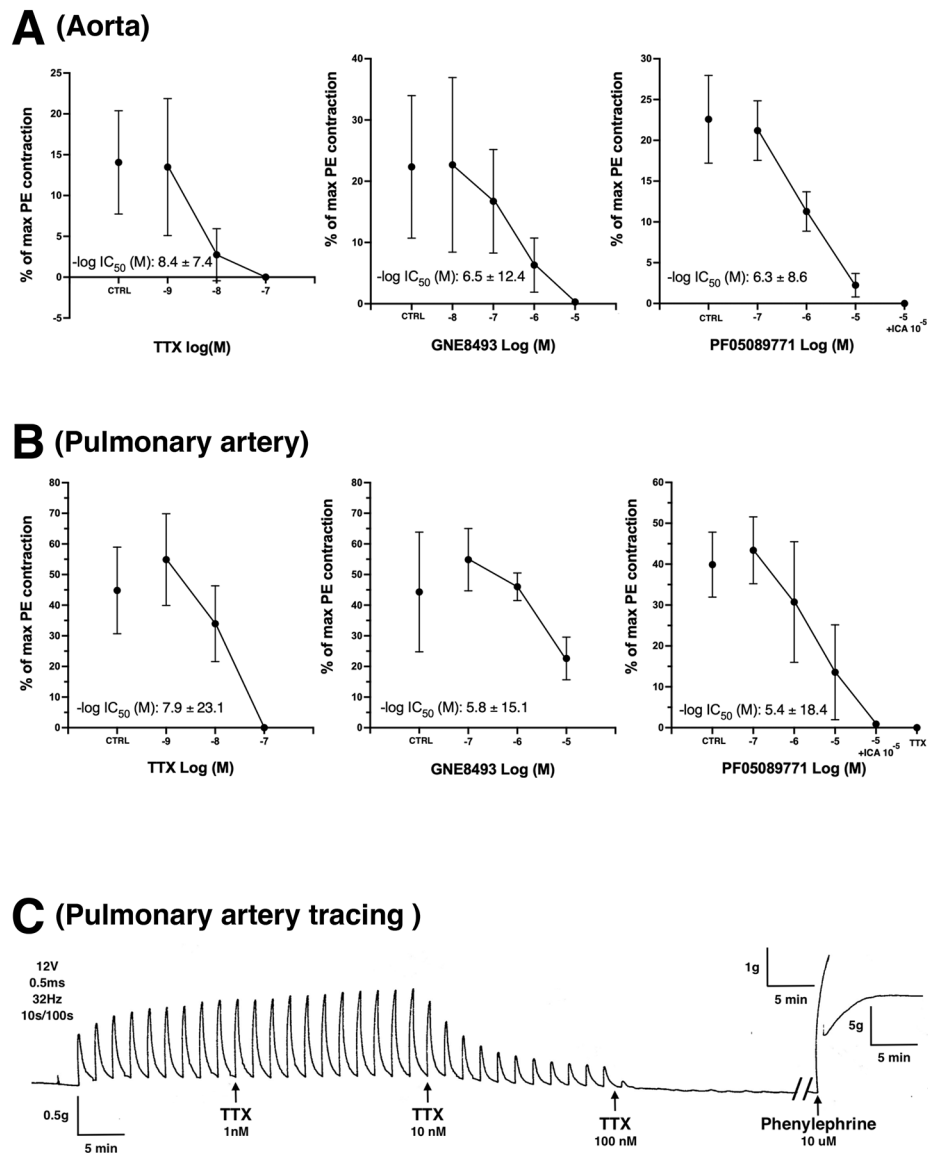


Figure 5. Concentration-inhibition curves for TTX-, GNE8483-, and PF05089771-inhibition of EFS-induced adrenergic contractions of (A) the guinea pig abdominal aorta ($n=5$, $n=7$, $n=5$, respectively) and (B) the guinea pig isolated pulmonary artery ($n=9$, $n=4$, $n=12$, respectively). (C) a representative tracing of guinea pig pulmonary artery EFS delivered for 10 s every 100 sec in the absence and presence of TTX; phenylephrine is given at the end of the experiment (note that the sensitivity is decreased before and during the phenylephrine-induced contraction). The nerve evoked contractions in the concentration-response curves are expressed as a % of the maximum phenylephrine (PE)-induced response. The data represent the mean $\pm$ SD of 6 different guinea pig experiments.

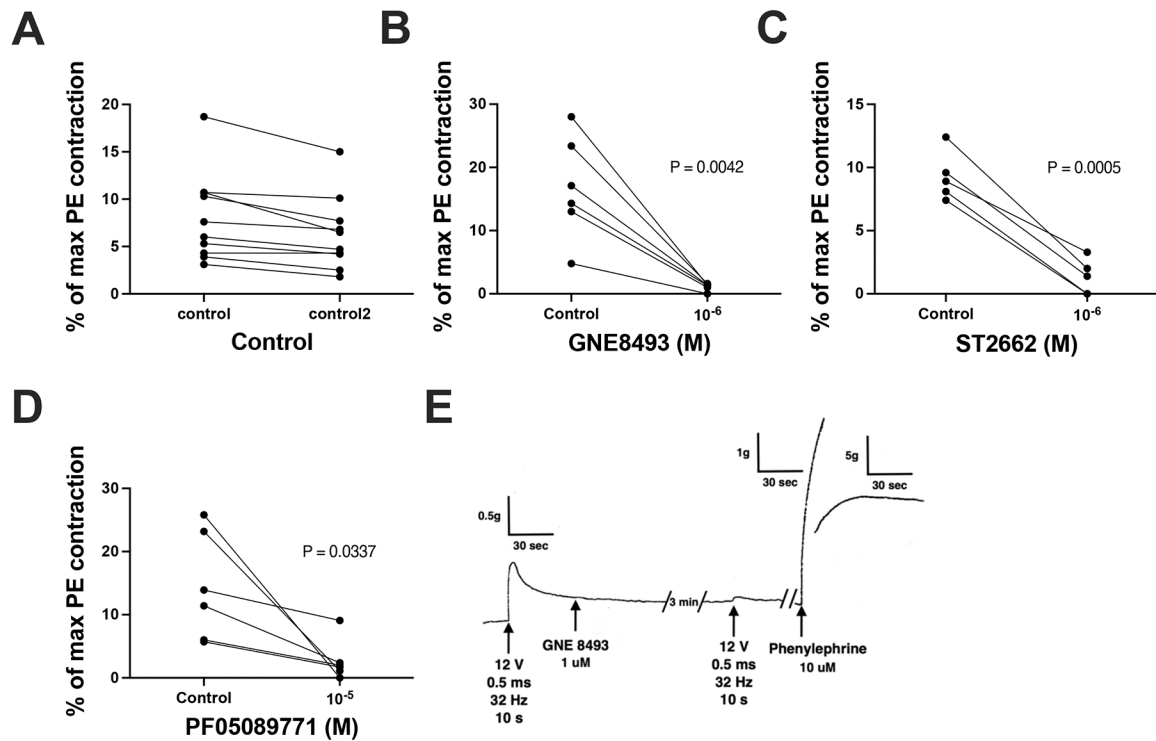


Figure 6.

The effect of (A) Vehicle control (n=10), (B) GNE8494 (n=6), (C) ST2662 (n=5), and (D) PF05089771 (n=6) on adrenergic contractions of human pulmonary arteries evoked by EFS. (E) an example of the response to a human pulmonary artery before and after GNE8483. The EFS-induced contractions are expressed as a % of the maximal adrenergic contractions taken as that evoked by 10 μ M phenylephrine (PE) added at the end of the experiment. P-value on the graph shows a statistical difference between the control and drug-treated values. The p-value for the differences between the delta of Control 1 vs. Control 2 values in the time controls compared to the delta between Control and tissues treated with GNE8493, ST2662, and P771 were 0.0005, 0.049, and 0.025, respectively.

Table 1.

Single-neuron RT-PCR primer sequences

Gene	Primer	Sequence	Product Size (bp)
β -actin	Forward (5'-3')	CCACCATGTACCCAGGCATT	189
	Reverse (3'-5')	CCGACTCATCGTACTCCTGC	
Na _v 1.1	Forward (5'-3')	AAGGGTTTCGCTTCTCCATT	206
	Reverse (3'-5')	GTCCTCAAAGGTGCTGTGCT	
Na _v 1.2	Forward (5'-3')	GAGACTTCAGTGGTGCTGGTG	265
	Reverse (3'-5')	AGCAGAGACTGGTGTGGAGAA	
Na _v 1.3	Forward (5'-3')	GTTTGTGGACCTGGGCAATG	121
	Reverse (3'-5')	CTTCACCGACTGGATCAGGG	
Na _v 1.6	Forward (5'-3')	CCTCCTATGGACGAAAAGACA	222
	Reverse (3'-5')	TGCAGATGGTGATAGCCAAG	
Na _v 1.7	Forward (5'-3')	GAGGAAAAGGGAGATGATGAGA	158
	Reverse (3'-5')	AACAGGGAGCCACGAATG	

Table 2.
Relative Selectivity of Na_v1 Blockers Used in This Study

	-log IC50 (M)								
	Na _v 1.1	Na _v 1.2	Na _v 1.3	Na _v 1.4	Na _v 1.5	Na _v 1.6	Na _v 1.7	Na _v 1.8	Na _v 1.9
Tetrodotoxin ^a	8.4	7.8	8.3	8.2	6.0	8.6	8.7	4.2	4.4
PF-05089771 ^b	6.1	6.9	5.0	5.0	4.6	6.8	8.0	<4.0	
ST-2262 ^c	<4.0	<4.0	4.2	4.1	<4.0	4.7	7.1	<4.0	
ICA121-431 ^d	7.6	6.6	7.9	<5.0	<4.5	4.9	5.0	<4.0	
GNE8493 ^e	5.1	5.3	5.1	5.0	<4.5	5.1	8.8	<4.0	<4.0

The estimated potencies of the sodium channel blockers were obtained from previously published studies. The studies were not all carried out using identical protocols, so one must be cautious when comparing between the compounds.

^aData obtained from (Alexander et al., 2023)
^bData obtained from (Alexandrou et al., 2016)
^cData obtained from (Pajouhesh et al., 2020)
^dData obtained from (McCormack et al., 2013)
^eData obtained from Dr. David Hackos at Genentech (see methods).